1 System Description

Consider a multi-period power–computing coordinated system over a scheduling horizon $\mathcal{T}$. The electrical layer is based on the IEEE 33-bus distribution network, which contains thermal units, wind power units, and photovoltaic (PV) units. Multiple geographically distributed data centers are connected to different buses of the distribution system. Each data center contains multiple computing nodes that can be turned on or off according to workload demand.

Two types of workloads are considered:

- **Interactive workload**: latency-sensitive requests generated by different source regions. Each request flow must be assigned to one feasible data center in each time period.
- **Batch workload**: delay-tolerant jobs with release times and deadlines. Batch jobs can be processed and migrated among data centers over time.

The system operates in a **grid-connected mode**. Bus 1 is treated as the root/slack bus through which the distribution network exchanges power with the upstream grid. The objective is to jointly optimize thermal generation dispatch, renewable power utilization, power imported from the upstream grid, data center activation, node-level workload allocation, and workload migration while satisfying the operating constraints of both the power network and the computing system.

2 Objective Function

The objective is to minimize the total operating cost:

$$\min \quad C^{\text{th}} + C^{\text{grid}} + C^{\text{sw}} + C^{\text{mig,I}} + C^{\text{mig,B}} + C^{\text{cur}} + C^{\text{late}}. \tag{1}$$

2.1 Thermal Generation Cost

The thermal unit model is simplified by retaining only the on/off status and the active power output:

$$C^{\text{th}} = \sum_{t\in\mathcal{T}} \sum_{g\in\mathcal{G}^{\text{th}}} \left(a_g \left(P^{\text{th}}_{g,t}\right)^2 + b_g P^{\text{th}}_{g,t} + c_g u_{g,t} \right). \tag{2}$$

2.2 Upstream Grid Purchase Cost

$$C^{\text{grid}} = \sum_{t\in\mathcal{T}} \pi^{\text{grid}}_t P^{\text{grid}}_t. \tag{3}$$

2.3 Node Switching Cost

$$C^{\mathrm{sw}} = \sum_{d\in\mathcal{D}} \sum_{n\in\mathcal{N}_d} \pi^{\mathrm{sw}}_{d,n}\eta_{d,n,1} + \sum_{t\in\mathcal{T}\setminus\{1\}} \sum_{d\in\mathcal{D}} \sum_{n\in\mathcal{N}_d} \pi^{\mathrm{sw}}_{d,n}\eta_{d,n,t}. \tag{4}$$

2.4 Interactive Workload Switching Cost

$$C^{\mathrm{mig,I}} = \sum_{t\in\mathcal{T}\setminus\{1\}} \sum_{r\in\mathcal{R}} \sum_{\substack{d,d'\in\mathcal{D}\\ d\neq d'}} \pi^{\mathrm{mig,I}}_{r} m^{\mathrm{I}}_{r,d,d',t}. \tag{5}$$

2.5 Batch Job Migration Cost

$$C^{\mathrm{mig,B}} = \sum_{t\in\mathcal{T}\setminus\{1\}} \sum_{j\in\mathcal{J}} \sum_{\substack{d,d'\in\mathcal{D}\\ d\neq d'}} \pi^{\mathrm{mig}}_{j} m^{\mathrm{B}}_{j,d,d',t}. \tag{6}$$

2.6 Renewable Curtailment Cost

$$C^{\mathrm{cur}} = \sum_{t\in\mathcal{T}} \left[\sum_{g\in\mathcal{G}^{\mathrm{w}}} \pi^{\mathrm{cur,w}}_{g} P^{\mathrm{cur,w}}_{g,t} + \sum_{g\in\mathcal{G}^{\mathrm{pv}}} \pi^{\mathrm{cur,pv}}_{g} P^{\mathrm{cur,pv}}_{g,t} \right]. \tag{7}$$

2.7 Batch Deadline Penalty

$$C^{\mathrm{late}} = \sum_{j\in\mathcal{J}} \pi^{\mathrm{late}}_{j}(1-\delta_j). \tag{8}$$

3 Constraints

3.1 Upstream Grid Exchange Constraints

$$0 \le P^{\mathrm{grid}}_{t} \le \overline{P}^{\mathrm{grid}}, \qquad \forall t\in\mathcal{T}. \tag{9}$$

$$\underline{Q}^{\mathrm{grid}} \le Q^{\mathrm{grid}}_{t} \le \overline{Q}^{\mathrm{grid}}, \qquad \forall t\in\mathcal{T}. \tag{10}$$

3.2 Simplified Thermal Unit Constraints

Active power output limits

$$P^{\min}_{g} u_{g,t} \le P^{\mathrm{th}}_{g,t} \le P^{\max}_{g} u_{g,t}, \qquad \forall g\in\mathcal{G}^{\mathrm{th}},\ \forall t\in\mathcal{T}. \tag{11}$$

Reactive power output limits

$$Q^{\min}_{g} u_{g,t} \le Q^{\mathrm{th}}_{g,t} \le Q^{\max}_{g} u_{g,t}, \qquad \forall g\in\mathcal{G}^{\mathrm{th}},\ \forall t\in\mathcal{T}. \tag{12}$$

Ramping constraints For the first period:

$$-RD_g \leq P^{\mathrm{th}}_{g,1} - P^{\mathrm{ini}}_{g,0} \leq RU_g, \qquad \forall g \in \mathcal{G}^{\mathrm{th}}. \tag{13}$$

For the remaining periods:

$$-RD_g \leq P^{\mathrm{th}}_{g,t} - P^{\mathrm{th}}_{g,t-1} \leq RU_g, \qquad \forall g \in \mathcal{G}^{\mathrm{th}},\ \forall t \in \mathcal{T} \setminus \{1\}. \tag{14}$$

3.3 Renewable Generation Constraints

Wind power constraints

$$0 \leq P^{\mathrm{w}}_{g,t} \leq \overline{P}^{\mathrm{w}}_{g,t}, \qquad \forall g \in \mathcal{G}^{\mathrm{w}},\ \forall t \in \mathcal{T}, \tag{15}$$

$$P^{\mathrm{w}}_{g,t} + P^{\mathrm{cur,w}}_{g,t} = \overline{P}^{\mathrm{w}}_{g,t}, \qquad \forall g \in \mathcal{G}^{\mathrm{w}},\ \forall t \in \mathcal{T}, \tag{16}$$

$$Q^{\mathrm{min,w}}_{g} \leq Q^{\mathrm{w}}_{g,t} \leq Q^{\mathrm{max,w}}_{g}, \qquad \forall g \in \mathcal{G}^{\mathrm{w}},\ \forall t \in \mathcal{T}. \tag{17}$$

PV power constraints

$$0 \leq P^{\mathrm{pv}}_{g,t} \leq \overline{P}^{\mathrm{pv}}_{g,t}, \qquad \forall g \in \mathcal{G}^{\mathrm{pv}},\ \forall t \in \mathcal{T}, \tag{18}$$

$$P^{\mathrm{pv}}_{g,t} + P^{\mathrm{cur,pv}}_{g,t} = \overline{P}^{\mathrm{pv}}_{g,t}, \qquad \forall g \in \mathcal{G}^{\mathrm{pv}},\ \forall t \in \mathcal{T}, \tag{19}$$

$$Q^{\mathrm{min,pv}}_{g} \leq Q^{\mathrm{pv}}_{g,t} \leq Q^{\mathrm{max,pv}}_{g}, \qquad \forall g \in \mathcal{G}^{\mathrm{pv}},\ \forall t \in \mathcal{T}. \tag{20}$$

3.4 Interactive Workload Constraints

Unique assignment over latency-feasible data centers

$$\sum_{d \in \mathcal{D}^{\mathrm{lat}}_r} x^{\mathrm{I}}_{r,d,t} = 1, \qquad \forall r \in \mathcal{R},\ \forall t \in \mathcal{T}. \tag{21}$$

Infeasible assignments are forbidden

$$x^{\mathrm{I}}_{r,d,t} = 0, \qquad \forall r \in \mathcal{R},\ \forall d \in \mathcal{D} \setminus \mathcal{D}^{\mathrm{lat}}_r,\ \forall t \in \mathcal{T}. \tag{22}$$

Interactive workload received by each data center

$$L^{\mathrm{I}}_{d,t} = \sum_{r \in \mathcal{R}} \phi^{\mathrm{I}}_r \lambda_{r,t} x^{\mathrm{I}}_{r,d,t}, \qquad \forall d \in \mathcal{D},\ \forall t \in \mathcal{T}. \tag{23}$$

Interactive assignment switching

$$m^{\mathrm{I}}_{r,d,d',t} \geq x^{\mathrm{I}}_{r,d,t-1} + x^{\mathrm{I}}_{r,d',t} - 1, \qquad \forall r \in \mathcal{R},\ \forall d \neq d',\ \forall t \in \mathcal{T} \setminus \{1\}. \tag{24}$$

3.5 Batch Workload Constraints

Location uniqueness within the valid time window

$$\sum_{d\in\mathcal{D}} z_{j,d,t} = 1, \qquad \forall j \in \mathcal{J},\ \forall t \in \mathcal{T}_j. \tag{25}$$

No location before release time or after deadline

$$z_{j,d,t} = 0, \qquad \forall j \in \mathcal{J},\ \forall d \in \mathcal{D},\ \forall t \in \mathcal{T} \setminus \mathcal{T}_j. \tag{26}$$

Execution only at the assigned data center

$$x^{\mathrm{B}}_{j,d,n,t} \le z_{j,d,t}, \qquad \forall j \in \mathcal{J},\ \forall d \in \mathcal{D},\ \forall n \in \mathcal{N}_d,\ \forall t \in \mathcal{T}. \tag{27}$$

Single-node execution per time period

$$\sum_{d\in\mathcal{D}} \sum_{n\in\mathcal{N}_d} x^{\mathrm{B}}_{j,d,n,t} \le 1, \qquad \forall j \in \mathcal{J},\ \forall t \in \mathcal{T}. \tag{28}$$

Processing amount constraints

$$0 \le f_{j,d,n,t} \le C^{\mathrm{cpu}}_{d,n} x^{\mathrm{B}}_{j,d,n,t}, \qquad \forall j \in \mathcal{J},\ \forall d \in \mathcal{D},\ \forall n \in \mathcal{N}_d,\ \forall t \in \mathcal{T}. \tag{29}$$

No execution before release time or after deadline

$$x^{\mathrm{B}}_{j,d,n,t} = 0, \quad f_{j,d,n,t} = 0, \qquad \forall j \in \mathcal{J},\ \forall d \in \mathcal{D},\ \forall n \in \mathcal{N}_d,\ \forall t \in \mathcal{T} \setminus \mathcal{T}_j. \tag{30}$$

Completion constraint

$$\sum_{t\in\mathcal{T}_j} \sum_{d\in\mathcal{D}} \sum_{n\in\mathcal{N}_d} f_{j,d,n,t} \ge W_j \delta_j, \qquad \forall j \in \mathcal{J}. \tag{31}$$

Batch migration constraint

$$m^{\mathrm{B}}_{j,d,d',t} \ge z_{j,d,t-1} + z_{j,d',t} - 1, \qquad \forall j \in \mathcal{J},\ \forall d \ne d',\ \forall t \in \mathcal{T} \setminus \{1\}. \tag{32}$$

3.6 Data Center Activation and Capacity Constraints

Node activation implies data center activation

$$s_{d,n,t} \le y_{d,t}, \qquad \forall d \in \mathcal{D},\ \forall n \in \mathcal{N}_d,\ \forall t \in \mathcal{T}. \tag{33}$$

Activation consistency

$$\sum_{n\in\mathcal{N}_d} s_{d,n,t} \le |\mathcal{N}_d| y_{d,t}, \qquad \forall d \in \mathcal{D},\ \forall t \in \mathcal{T}. \tag{34}$$

$$y_{d,t} \le \sum_{n\in\mathcal{N}_d} s_{d,n,t}, \qquad \forall d \in \mathcal{D},\ \forall t \in \mathcal{T}. \tag{35}$$

Batch jobs can only be executed on active nodes

$$x^{\mathrm{B}}_{j,d,n,t} \le s_{d,n,t}, \qquad \forall j \in \mathcal{J},\ \forall d \in \mathcal{D},\ \forall n \in \mathcal{N}_d,\ \forall t \in \mathcal{T}. \tag{36}$$

Joint computing capacity constraint

$$L^{\mathrm{I}}_{d,t} + \sum_{j\in\mathcal{J}} \sum_{n\in\mathcal{N}_d} f_{j,d,n,t} \le \sum_{n\in\mathcal{N}_d} C^{\mathrm{cpu}}_{d,n} s_{d,n,t}, \qquad \forall d \in \mathcal{D},\ \forall t \in \mathcal{T}. \tag{37}$$

3.7 Data Center Power Consumption Model

IT power consumption

$$P^{\mathrm{IT}}_{d,t} = \sum_{n\in\mathcal{N}_d} P^{\mathrm{idle}}_{d,n} s_{d,n,t} + \alpha_d \left(L^{\mathrm{I}}_{d,t} + \sum_{j\in\mathcal{J}} \sum_{n\in\mathcal{N}_d} f_{j,d,n,t} \right), \qquad \forall d \in \mathcal{D},\ \forall t \in \mathcal{T}. \tag{38}$$

Total data center power consumption

$$P^{\mathrm{DC}}_{d,t} = \mathrm{PUE}_d P^{\mathrm{IT}}_{d,t}, \qquad \forall d \in \mathcal{D},\ \forall t \in \mathcal{T}. \tag{39}$$

Reactive power consumption

$$Q^{\mathrm{DC}}_{d,t} = P^{\mathrm{DC}}_{d,t} \tan(\arccos(\mathrm{pf}_d)), \qquad \forall d \in \mathcal{D},\ \forall t \in \mathcal{T}. \tag{40}$$

3.8 Node Switching Cost Linearization

For the first period:

$$\eta_{d,n,1} \ge s_{d,n,1} - s^{\mathrm{ini}}_{d,n,0}, \qquad \forall d \in \mathcal{D},\ \forall n \in \mathcal{N}_d, \tag{41}$$

$$\eta_{d,n,1} \ge s^{\mathrm{ini}}_{d,n,0} - s_{d,n,1}, \qquad \forall d \in \mathcal{D},\ \forall n \in \mathcal{N}_d. \tag{42}$$

For the remaining periods:

$$\eta_{d,n,t} \geq s_{d,n,t} - s_{d,n,t-1}, \qquad \forall d \in \mathcal{D},\ \forall n \in \mathcal{N}_d,\ \forall t \in \mathcal{T} \setminus \{1\}, \tag{43}$$

$$\eta_{d,n,t} \geq s_{d,n,t-1} - s_{d,n,t}, \qquad \forall d \in \mathcal{D},\ \forall n \in \mathcal{N}_d,\ \forall t \in \mathcal{T} \setminus \{1\}, \tag{44}$$

$$\eta_{d,n,t} \geq 0, \qquad \forall d \in \mathcal{D},\ \forall n \in \mathcal{N}_d,\ \forall t \in \mathcal{T}. \tag{45}$$

3.9 Distribution Network Constraints

The total active and reactive demands at bus i are defined as

$$P_{i,t}^{\text{load,tot}} = P_{i,t}^{L} + \sum_{d \in \mathcal{D}: b(d)=i} P_{d,t}^{\text{DC}}, \qquad \forall i \in \mathcal{B},\ \forall t \in \mathcal{T}, \tag{46}$$

$$Q_{i,t}^{\text{load,tot}} = Q_{i,t}^{L} + \sum_{d \in \mathcal{D}: b(d)=i} Q_{d,t}^{\text{DC}}, \qquad \forall i \in \mathcal{B},\ \forall t \in \mathcal{T}. \tag{47}$$

The total active generation at bus 1 is defined as

$$P_{1,t}^{\text{gen}} = P_{t}^{\text{grid}} + \sum_{g \in \mathcal{G}_1^{\text{th}}} P_{g,t}^{\text{th}} + \sum_{g \in \mathcal{G}_1^{\text{w}}} P_{g,t}^{\text{w}} + \sum_{g \in \mathcal{G}_1^{\text{pv}}} P_{g,t}^{\text{pv}}, \qquad \forall t \in \mathcal{T}. \tag{48}$$

The total reactive generation at bus 1 is defined as

$$Q_{1,t}^{\text{gen}} = Q_{t}^{\text{grid}} + \sum_{g \in \mathcal{G}_1^{\text{th}}} Q_{g,t}^{\text{th}} + \sum_{g \in \mathcal{G}_1^{\text{w}}} Q_{g,t}^{\text{w}} + \sum_{g \in \mathcal{G}_1^{\text{pv}}} Q_{g,t}^{\text{pv}}, \qquad \forall t \in \mathcal{T}. \tag{49}$$

For buses $i \in \mathcal{B} \setminus \{1\}$, the total active and reactive generation are

$$P_{i,t}^{\text{gen}} = \sum_{g \in \mathcal{G}_i^{\text{th}}} P_{g,t}^{\text{th}} + \sum_{g \in \mathcal{G}_i^{\text{w}}} P_{g,t}^{\text{w}} + \sum_{g \in \mathcal{G}_i^{\text{pv}}} P_{g,t}^{\text{pv}}, \qquad \forall i \in \mathcal{B} \setminus \{1\},\ \forall t \in \mathcal{T}, \tag{50}$$

$$Q_{i,t}^{\text{gen}} = \sum_{g \in \mathcal{G}_i^{\text{th}}} Q_{g,t}^{\text{th}} + \sum_{g \in \mathcal{G}_i^{\text{w}}} Q_{g,t}^{\text{w}} + \sum_{g \in \mathcal{G}_i^{\text{pv}}} Q_{g,t}^{\text{pv}}, \qquad \forall i \in \mathcal{B} \setminus \{1\},\ \forall t \in \mathcal{T}. \tag{51}$$

A branch-flow (DistFlow) model is adopted.

Active power balance

$$P_{ij,t} = \sum_{k:(j,k) \in \mathcal{L}} P_{jk,t} + P_{j,t}^{\text{load,tot}} - P_{j,t}^{\text{gen}} + r_{ij} \ell_{ij,t}, \qquad \forall (i,j) \in \mathcal{L},\ \forall t \in \mathcal{T}. \tag{52}$$

Reactive power balance

$$Q_{ij,t} = \sum_{k:(j,k) \in \mathcal{L}} Q_{jk,t} + Q_{j,t}^{\text{load,tot}} - Q_{j,t}^{\text{gen}} + x_{ij} \ell_{ij,t}, \qquad \forall (i,j) \in \mathcal{L},\ \forall t \in \mathcal{T}. \tag{53}$$

Voltage drop equation

$$V_{j,t} = V_{i,t} - 2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) + (r_{ij}^2 + x_{ij}^2)\ell_{ij,t}, \qquad \forall (i,j) \in \mathcal{L},\ \forall t \in \mathcal{T}. \tag{54}$$

Second-order cone relaxation

$$P_{ij,t}^2 + Q_{ij,t}^2 \le V_{i,t}\ell_{ij,t}, \qquad \forall (i,j) \in \mathcal{L},\ \forall t \in \mathcal{T}. \tag{55}$$

Voltage limits

$$\underline{V}_i \le V_{i,t} \le \overline{V}_i, \qquad \forall i \in \mathcal{B},\ \forall t \in \mathcal{T}. \tag{56}$$

Line flow limits

$$P_{ij,t}^2 + Q_{ij,t}^2 \le \overline{S}_{ij}^2, \qquad \forall (i,j) \in \mathcal{L},\ \forall t \in \mathcal{T}. \tag{57}$$

Slack bus voltage

$$V_{1,t} = V^{\text{slack}}, \qquad \forall t \in \mathcal{T}. \tag{58}$$

4 Variable Domains

$$\begin{aligned}
u_{g,t} &\in \{0,1\}, && \forall g \in \mathcal{G}^{\text{th}},\ \forall t \in \mathcal{T}, && (59)\\
s_{d,n,t}, y_{d,t} &\in \{0,1\}, && \forall d \in \mathcal{D},\ \forall n \in \mathcal{N}_d,\ \forall t \in \mathcal{T}, && (60)\\
x_{r,d,t}^{\text{I}} &\in \{0,1\}, && \forall r \in \mathcal{R},\ \forall d \in \mathcal{D},\ \forall t \in \mathcal{T}, && (61)\\
m_{r,d,d',t}^{\text{I}} &\in \{0,1\}, && \forall r \in \mathcal{R},\ \forall d \neq d',\ \forall t \in \mathcal{T} \setminus \{1\}, && (62)\\
z_{j,d,t}, x_{j,d,n,t}^{\text{B}}, m_{j,d,d',t}^{\text{B}}, \delta_j &\in \{0,1\}, && \forall j \in \mathcal{J},\ \forall d, d' \in \mathcal{D},\ \forall n \in \mathcal{N}_d,\ \forall t \in \mathcal{T}, && (63)
\end{aligned}$$

The remaining variables are continuous, with nonnegativity imposed where needed.

5 Compact Formulation

The overall power–computing coordinated scheduling problem can be summarized as follows:

$$\begin{aligned}
\min \quad & C^{\text{th}} + C^{\text{grid}} + C^{\text{sw}} + C^{\text{mig,I}} + C^{\text{mig,B}} + C^{\text{cur}} + C^{\text{late}} \\
\text{s.t.} \quad & \text{constraints } (9) - (63).
\end{aligned} \tag{64}$$

6 Model Characteristics

The proposed model has the following characteristics:

- It explicitly couples the power system and the computing system through the data center power consumption model.
- It operates in a grid-connected mode by introducing upstream grid power injection at the root bus.
- It distinguishes interactive and batch workloads by using different routing, service, and migration constraints.
- It includes multiple binary variables for thermal unit commitment, data center activation, node switching, interactive workload assignment, and batch job placement/migration.
- The thermal generation cost is modeled as a quadratic function of active power output.
- With the branch-flow network model and second-order cone constraints, the problem can be solved as an MIQCP/MISOCP model.

7 Remarks

1. The reactive power capability of thermal, wind, and PV units is approximated by independent box constraints for tractability.
2. If reactive power optimization is not needed, all reactive power variables and constraints can be removed to obtain a simpler active-power-only model.
3. If all batch jobs must be completed, δ_j can be fixed to 1 for all $j \in \mathcal{J}$.
4. If parallel execution of batch jobs is allowed, constraint (28) can be removed.
5. If interactive workload splitting among multiple data centers is needed, $x^{\mathrm{I}}_{r,d,t}$ can be relaxed from binary variables to continuous assignment ratios.